Exercise 01 – Solution Summary

Notation:

- $\wedge$ = AND, $\vee$ = OR, $\overline{A}$ = NOT A
- ${}^{a}Q$ = old state, ${}^{n}Q$ = new state (from German *alt* / *neu*).

Task 1.1

Idea: Write the target function as a Boolean expression in **DNF**, then rewrite it so that it contains **only NAND operations** $\overline{X \wedge Y}$. Each NAND operation is one gate.

- **DNF (Disjunctive Normal Form)** is an OR of AND terms, e.g. $(\overline{A} \wedge B) \vee (A \wedge \overline{B})$.

1. **NAND gate**
   - Boolean equation: Out $= \overline{A \wedge B}$
   - The output is 0 only if both inputs are 1.

A	B	Out
0	0	1
0	1	1
1	0	1
1	1	0

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2. **Building blocks: NOT, AND and OR using only NAND gates**
   - **NOT**: connect both inputs of a NAND gate together (idempotent law $A \wedge A = A$):

$$\overline{A} = \overline{A \wedge A}$$

   - **AND**: a NAND followed by a NOT (double negation law $A = \overline{\overline{A}}$):

$$A \wedge B = \overline{\overline{A \wedge B}} = \overline{\overline{A \wedge B} \wedge \overline{A \wedge B}}$$

   - **OR**: invert both inputs, then NAND (De Morgan's law $\overline{A \vee B} = \overline{A} \wedge \overline{B}$):

$$A \vee B = \overline{\overline{A \vee B}} = \overline{\overline{A} \wedge \overline{B}} = \overline{\overline{A \wedge A} \wedge \overline{B \wedge B}}$$

   Note: Since NOT, AND and OR can all be built from NAND, NAND is called a **universal gate**. The same holds for NOR.

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3. **XOR gate**
   - The output is 1 only if the two inputs are different:

A	B	XOR
0	0	0
0	1	1
1	0	1
1	1	0

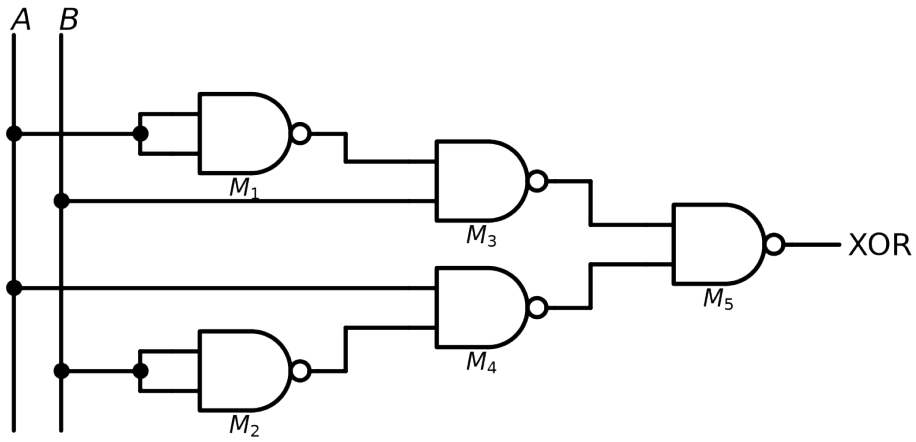
- The rows with output 1 are $AB = 01$ and $AB = 10$, so the DNF is:

$$\text{Out} = (\overline{A} \wedge B) \vee (A \wedge \overline{B})$$

- Apply double negation, then De Morgan's law to the inner negation:

$$\begin{aligned}\text{Out} &= \overline{\overline{\overline{\overline{(\overline{A} \wedge B) \vee (A \wedge \overline{B})}}}} \\ &= \overline{\overline{\overline{A} \wedge B} \wedge \overline{A \wedge \overline{B}}} \\ &= \overline{\overline{\overline{A} \wedge \overline{A} \wedge B} \wedge \overline{A \wedge \overline{B} \wedge B}}\end{aligned}$$

- Count the NAND gates: 2 inverters ($\overline{A \wedge A}$, $\overline{B \wedge B}$) + 2 NANDs + 1 output NAND = **5 gates**



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4. **Optimization: XOR with only 4 NAND gates**

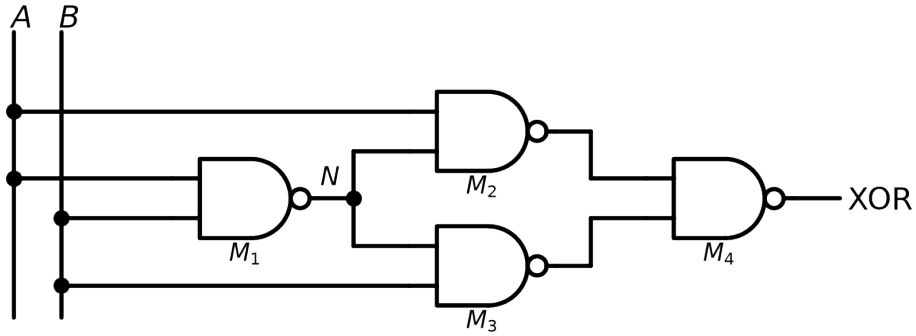
- Let $N = \overline{A \wedge B}$. Then:

$$\begin{aligned}A \wedge N &= A \wedge (\overline{A} \vee \overline{B}) = (A \wedge \overline{A}) \vee (A \wedge \overline{B}) = A \wedge \overline{B} \\ B \wedge N &= B \wedge (\overline{A} \vee \overline{B}) = \overline{A} \wedge B\end{aligned}$$

- So N can replace **both** inverters:

$$\text{Out} = \overline{\overline{A \wedge N} \wedge \overline{B \wedge N}}, \qquad N = \overline{A \wedge B}$$

- Gates: N, $\overline{A \wedge N}$, $\overline{B \wedge N}$ and the output NAND = **4 gates**.

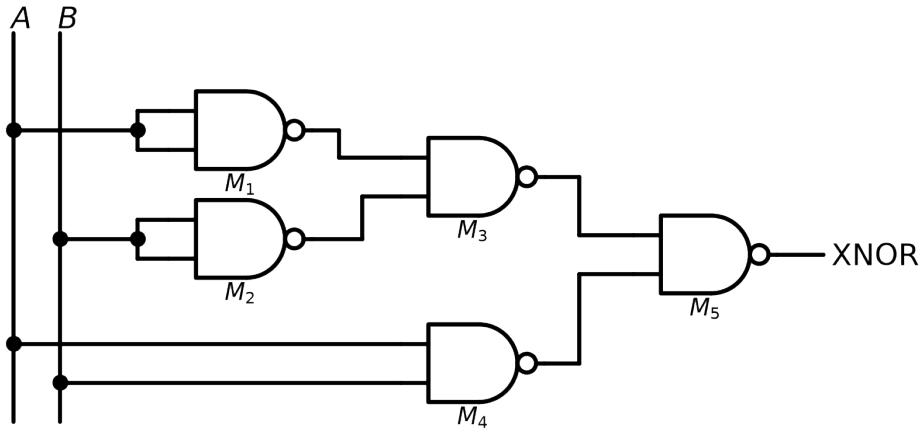


5. **XNOR gate**

- XNOR is the negation of XOR. The output is 1 only if the two inputs are equal, i.e. the rows $AB = 00$ and $AB = 11$:

$$\begin{aligned}\text{Out} &= (\overline{A} \wedge \overline{B}) \vee (A \wedge B) \\ &= \overline{\overline{(\overline{A} \wedge \overline{B}) \vee (A \wedge B)}} \\ &= \overline{\overline{\overline{A} \wedge \overline{B}} \wedge \overline{A \wedge B}} \\ &= \overline{\overline{\overline{A} \wedge \overline{A}} \wedge \overline{\overline{B} \wedge \overline{B}} \wedge \overline{A \wedge B}}\end{aligned}$$

- Count the NAND gates: 2 inverters ($\overline{A \wedge A}$, $\overline{B \wedge B}$) + 2 NANDs + 1 output NAND = **5 gates**.



- Alternatively, add a NOT gate after the optimized XOR: XNOR $= \overline{\text{XOR} \wedge \text{XOR}}$, same 5 gates.

6. **2-input Multiplexer (MUX)**

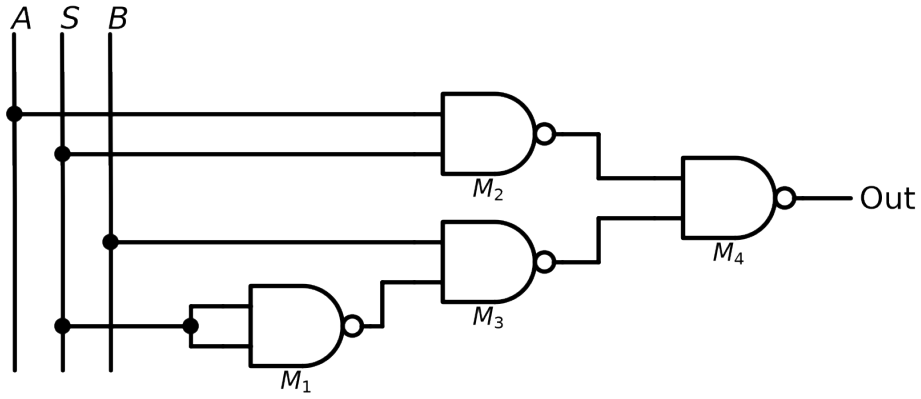
- S selects which input is passed to the output: $S = 1$ selects A, $S = 0$ selects B.

S	Out
0	B
1	A

- DNF and NAND form:

$$\begin{aligned}\text{Out} &= (A \wedge S) \vee (B \wedge \overline{S}) \\ &= \overline{\overline{A \wedge S} \wedge \overline{B \wedge \overline{S}}} \\ &= \overline{\overline{A \wedge S} \wedge \overline{B \wedge \overline{S \wedge S}}}\end{aligned}$$

- Gates: 1 inverter for $\overline{S}$ + 2 NANDs + 1 output NAND = **4 gates**.

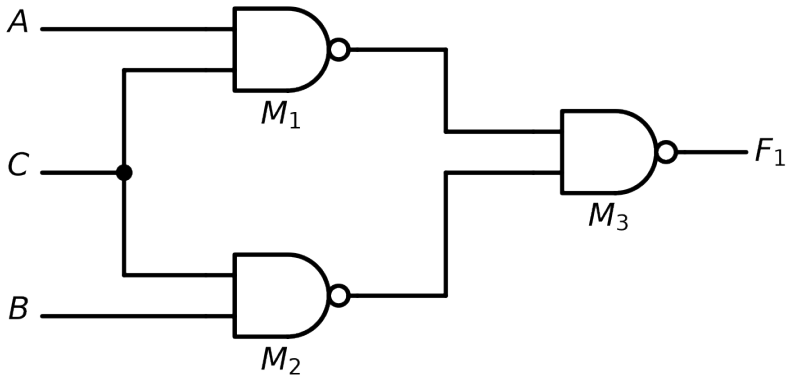


7. **Homework**: Try to design XOR, XNOR and the 2-input MUX **using only NOR gates**.

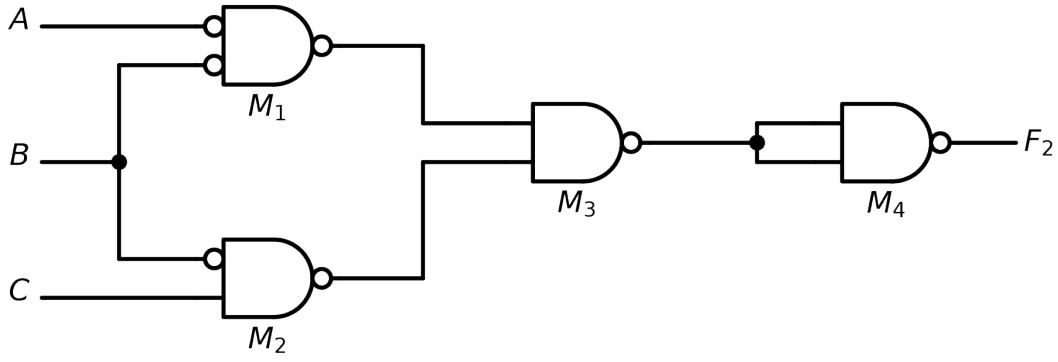
Task 1.2

Idea: Same method as Task 1.1. Inverted inputs are drawn as bubbles, as in the official solution.

1. $F_1 = (A \wedge C) \vee (B \wedge C) = \overline{\overline{A \wedge C} \wedge \overline{B \wedge C}}$



2. $F_2 = (A \vee B) \wedge (\overline{C} \vee B) = \overline{\overline{\overline{A} \wedge \overline{B}} \wedge \overline{C \wedge \overline{B}}}$



Task 1.3

Idea: To complement the expression, apply **De Morgan's law**. If possible, **simplify the expression first**. All laws are listed in the Appendix.

1. **Task (a):** $F = (\overline{A} \wedge \overline{B} \wedge C) \vee (\overline{A} \wedge B \wedge \overline{C}) \vee (A \wedge B \wedge \overline{C})$

   - **Step 1 – Simplify F.** The last two terms share $B \wedge \overline{C}$:

   $$\begin{aligned}(\overline{A} \wedge B \wedge \overline{C}) \vee (A \wedge B \wedge \overline{C}) &= (B \wedge \overline{C}) \wedge (\overline{A} \vee A) \\ &= (B \wedge \overline{C}) \wedge 1 \\ &= B \wedge \overline{C}\end{aligned}$$

   $$\Rightarrow \quad F = (\overline{A} \wedge \overline{B} \wedge C) \vee (B \wedge \overline{C})$$

   - **Step 2 – Complement with De Morgan's law.**:

   $$\overline{F} = \overline{\overline{A} \wedge \overline{B} \wedge C} \wedge \overline{B \wedge \overline{C}}$$

   - **Step 3 – De Morgan's law again inside each term.**:

   $$\overline{F} = (\overline{\overline{A}} \vee \overline{\overline{B}} \vee \overline{C}) \wedge (\overline{B} \vee \overline{\overline{C}})$$

   - **Step 4 – Double negation $\overline{\overline{A}} = A$:**

   $$\boxed{\overline{F} = (A \vee B \vee \overline{C}) \wedge (\overline{B} \vee C)}$$

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2. **Task (b):** $F = (A \wedge B \wedge C) \vee (\overline{A} \wedge \overline{B} \wedge C)$

$$\begin{aligned}\overline{F} &= \overline{(A \wedge B \wedge C) \vee (\overline{A} \wedge \overline{B} \wedge C)} \\ &= \overline{A \wedge B \wedge C} \wedge \overline{\overline{A} \wedge \overline{B} \wedge C} \\ &= (\overline{A} \vee \overline{B} \vee \overline{C}) \wedge (\overline{\overline{A}} \vee \overline{\overline{B}} \vee \overline{C}) \\ &= \boxed{(\overline{A} \vee \overline{B} \vee \overline{C}) \wedge (A \vee B \vee \overline{C})}\end{aligned}$$

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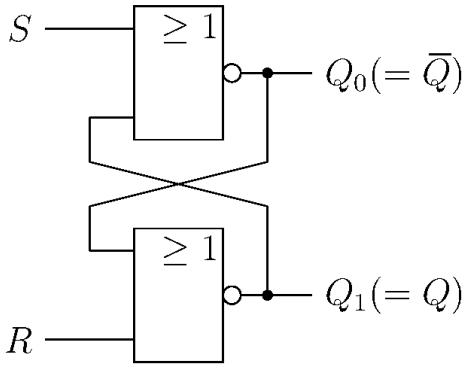
3. **Task (c):** $F = (A \wedge B \wedge C) \vee (A \wedge B \wedge \overline{C}) \vee (A \wedge \overline{B})$

$$\begin{aligned}F &= (A \wedge B) \wedge (C \vee \overline{C}) \vee (A \wedge \overline{B}) \\ &= (A \wedge B) \vee (A \wedge \overline{B}) \\ &= A \wedge (B \vee \overline{B}) \\ &= A\end{aligned}$$

$$\boxed{\overline{F} = \overline{A}}$$

Task 1.4

1. **Task (a):** The **RS latch** is a 1-bit memory built from two cross-coupled NOR gates. Its outputs are $Q_1 = Q$ and $Q_0 = \overline{Q}$. Because the latch is built from NOR gates, its inputs are **active-high** (S, R): $SR = 00$ stores, $S = 1$ sets and $R = 1$ resets.



A NOR gate outputs 0 as soon as one of its inputs is 1. From this we get:

S	R	Q	Function
1	0	1	Set
0	1	0	Reset
0	0	old Q	Store
1	1	0 (and $\overline{Q} = 0$)	**Forbidden**

Note: $SR = 11$ is forbidden because Q and $\overline{Q}$ are both 0, so the outputs are no longer complementary.

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2. **Task (b):** The **logic automaton graph** lists all output states and all possible state transitions, along with the input conditions required for each transition.

- To get the graph, first build the **full switching table**. Inputs are S, R and the old outputs ${}^aQ_1$, ${}^aQ_0$; outputs are the new outputs ${}^nQ_1$, ${}^nQ_0$. From the circuit:

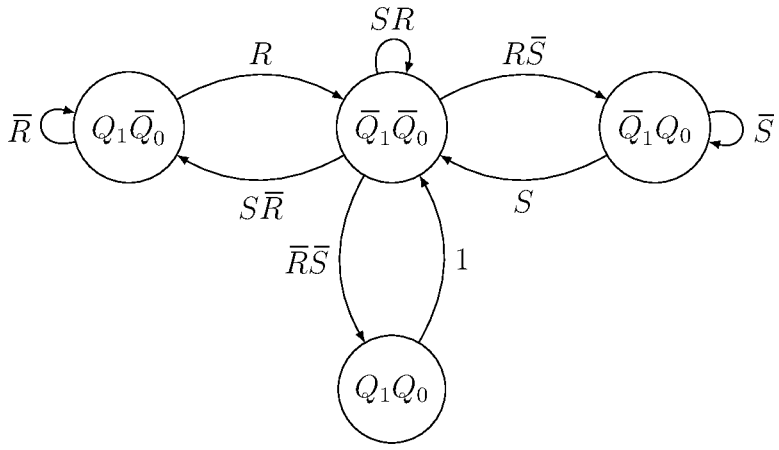
$$ {}^nQ_1 = \overline{R \vee {}^aQ_0}, \qquad {}^nQ_0 = \overline{S \vee {}^aQ_1} $$

${}^aQ_1$	${}^aQ_0$	S	R	${}^nQ_1$	${}^nQ_0$
0	0	0	0	1	1
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	0	0
0	1	0	0	0	1
0	1	0	1	0	1
0	1	1	0	0	0
0	1	1	1	0	0
1	0	0	0	1	0
1	0	0	1	0	0

${}^aQ_1$	${}^aQ_0$	S	R	${}^nQ_1$	${}^nQ_0$
1	0	1	0	1	0
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	0	0

- Each old state ${}^aQ_1\,{}^aQ_0$ (4 rows) becomes one circle. Rows with the same new state are merged into one arrow, and the arrow is labeled with the input condition:

Old state	Condition → new state
$\overline{Q_1}\,\overline{Q_0}$	$\overline{S}\,\overline{R} \to Q_1Q_0$, $\overline{S}R \to \overline{Q_1}Q_0$, $S\overline{R} \to Q_1\overline{Q_0}$, $SR \to \overline{Q_1}\,\overline{Q_0}$
$\overline{Q_1}Q_0$	$\overline{S} \to \overline{Q_1}Q_0$ (stay), $S \to \overline{Q_1}\,\overline{Q_0}$
$Q_1\overline{Q_0}$	$\overline{R} \to Q_1\overline{Q_0}$ (stay), $R \to \overline{Q_1}\,\overline{Q_0}$
Q_1Q_0	1 (any input) $\to \overline{Q_1}\,\overline{Q_0}$

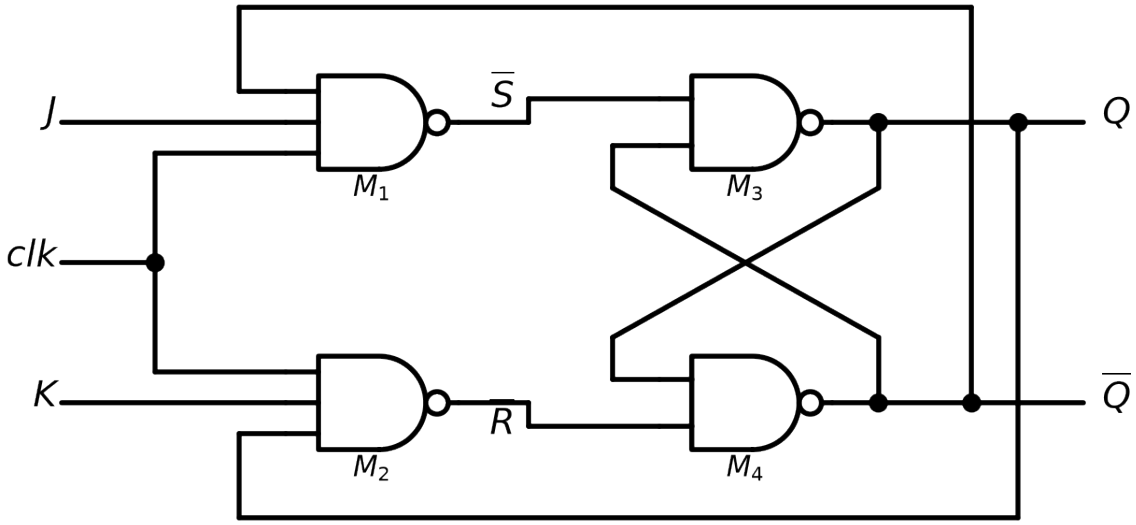


3. **Task (c):** The **reduced switching table** lists only the inputs S, R and the new state ${}^nQ$. It is the function table from Task (a), written with ${}^aQ$ / ${}^nQ$:

S	R	${}^nQ$	Comment
1	0	1	Set
0	1	0	Reset
0	0	${}^aQ$	Store, if ${}^aQ_1 = \overline{{}^aQ_0}$
1	1	0	Forbidden: ${}^nQ_1 = {}^nQ_0 = 0$

Task 1.5

1. **Circuit:** The JK flip-flop can be constructed using two 3-input NAND gates (M_1, M_2) and one RS latch. Because the latch is built from NAND gates (M_3, M_4), its inputs are **active-low** ($\overline{S}$, $\overline{R}$): $\overline{S}\,\overline{R} = 11$ stores, $\overline{S} = 0$ sets and $\overline{R} = 0$ resets.



$$\overline{S} = \overline{J \wedge clk \wedge \overline{Q}}, \qquad \overline{R} = \overline{K \wedge clk \wedge Q}$$

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2. **How it works**

- **clk = 0:** M_1 and M_2 output 1, so $\overline{S}\,\overline{R} = 11$ and the latch stores: ${}^{n}Q = {}^{a}Q$.
- **clk = 1:** M_1 and M_2 pass J and K to the latch, gated by the feedback:

$$\overline{S} = \overline{J \wedge {}^{a}\overline{Q}}, \qquad \overline{R} = \overline{K \wedge {}^{a}Q}$$

Note: As long as the latch is in a valid state, ${}^{a}Q$ and ${}^{a}\overline{Q}$ are complementary (never 1 at the same time). Therefore $\overline{S}$ and $\overline{R}$ can never both be 0, the latch never receives the forbidden input, and the new state is valid again.

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3. **Task (a): Switching table (excitation table)**

- There are only **3 independent inputs**: ${}^{a}Q$, J, K. (${}^{a}\overline{Q}$ is just the complement of ${}^{a}Q$.) This gives $2^3 = 8$ rows.
- For each row: compute $\overline{S}$, $\overline{R}$ from the equations above, then apply the latch: $\overline{S} = 0$ sets, $\overline{R} = 0$ resets, $\overline{S}\,\overline{R} = 11$ stores.

${}^{a}Q$	J	K	$\overline{S}$	$\overline{R}$	${}^{n}Q$	${}^{n}\overline{Q}$
0	0	0	1	1	0	1
0	0	1	1	1	0	1
0	1	0	0	1	1	0
0	1	1	0	1	1	0
1	0	0	1	1	1	0

${}^{a}Q$	J	K	$\overline{S}$	$\overline{R}$	${}^{n}Q$	${}^{n}\overline{Q}$
1	0	1	1	0	0	1
1	1	0	1	1	1	0
1	1	1	1	0	0	1

Note: The switching table assumes only **one** state change per clock pulse, i.e. clk = 1 lasts just long enough for the latch to switch once.

4. **Task (b): Reduced switching table**

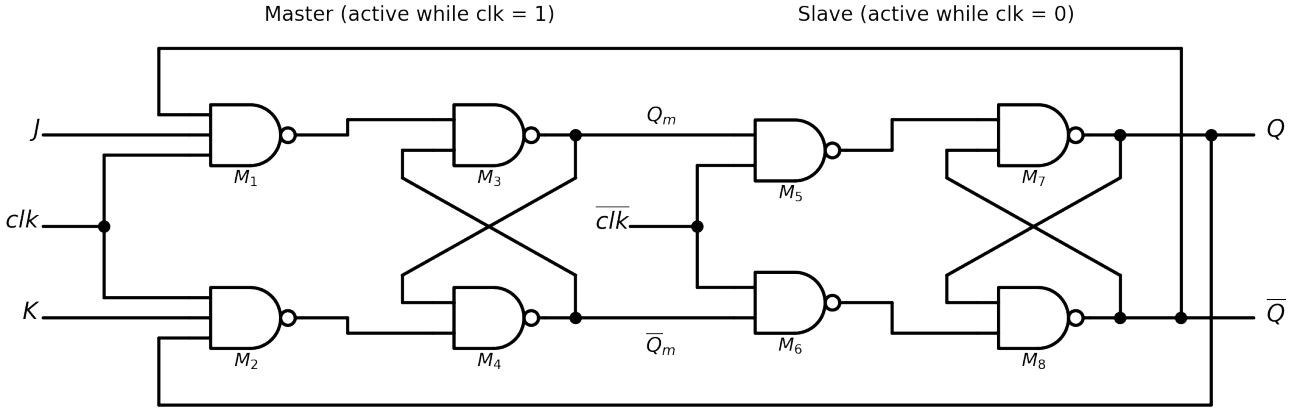
- Group the rows of the switching table by JK:

J	K	${}^{n}Q$	Comment
0	0	${}^{a}Q$	Store
0	1	0	Reset
1	0	1	Set
1	1	${}^{a}\overline{Q}$	Toggle

Note: Compared with the RS latch, the input combination $JK = 11$ is not forbidden: the JK flip-flop **toggles** its output.

5. **Supplement: master-slave JK flip-flop**

- The circuit in step 1 is **level-sensitive** (a gated JK latch): while clk = 1, a change of Q is fed back to M_1, M_2 immediately. With $J = K = 1$ the output would toggle again and again (*race-around*).
- A **master-slave** JK flip-flop adds a second latch:



clk	Master (M_1–M_4)	Slave (M_5–M_8)
1	takes over J, K	stores, Q unchanged
0	stores	copies Q_m to Q

Task 1.6

Idea: A constant current charges the capacitor, so its voltage rises **linearly**. The current follows directly from the capacitor equations.

1. Charge and current of a capacitor:

$$Q = C \cdot V \qquad \text{and} \qquad I = \frac{dQ}{dt}$$

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2. Since C is constant:

$$I = \frac{dQ}{dt} = C\frac{dV}{dt} = C\frac{\Delta V}{\Delta t}$$

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3. With $C = 25\,\text{fF}$, $\Delta V = 0.6\,\text{V} - 0\,\text{V} = 0.6\,\text{V}$ and $\Delta t = 30\,\text{ps}$:

$$I = \frac{25 \cdot 10^{-15}\,\frac{\text{As}}{\text{V}} \cdot 0.6\,\text{V}}{30 \cdot 10^{-12}\,\text{s}} = 0.5 \cdot 10^{-3}\,\text{A} = \boxed{0.5\,\text{mA}}$$

Task 1.7

Idea: Step response circuit, set up the differential equation with KVL, solve it for t, then insert the values.

1. Let V_{out} be the voltage across the capacitor, with $V_{out}(0) = 0$. The time constant is:

$$\tau = RC = 12.5\,\text{k}\Omega \cdot 100\,\text{fF} = 1.25\,\text{ns}$$

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2. Kirchhoff's voltage law and the capacitor current $I = C\,\frac{dV_{out}}{dt}$ give:

$$\begin{aligned} V_{in} &= IR + V_{out} = RC\,\frac{dV_{out}}{dt} + V_{out} \\ \Rightarrow \quad dt &= RC\,\frac{dV_{out}}{V_{in} - V_{out}} \end{aligned}$$

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3. Integrate both sides (integration variable v, from 0 to V_{out}):

$$\begin{aligned} \int_0^t dt' &= RC \int_0^{V_{out}} \frac{dv}{V_{in} - v} \\ t &= -RC\,\ln(V_{in} - v)\Big|_0^{V_{out}} \\ &= -\tau\,\ln\frac{V_{in} - V_{out}}{V_{in}} = -\tau\,\ln\left(1 - \frac{V_{out}}{V_{in}}\right) \end{aligned}$$

Solved for V_{out}, this is the well-known step response: $V_{out}(t) = V_{in}\left(1 - e^{-t/\tau}\right)$.

4. **Task (a):** $V_{out} = 0.6\,\text{V}$ (50 % of V_{in}):

$$t_a = -1.25\,\text{ns} \cdot \ln\left(1 - \frac{0.6\,\text{V}}{1.2\,\text{V}}\right) = \tau \ln 2 \approx \boxed{0.87\,\text{ns}}$$

5. **Task (b):** $V_{out} = 1.2\,\text{V}$ (100 % of V_{in}):

$$t_b = -1.25\,\text{ns} \cdot \ln\left(1 - \frac{1.2\,\text{V}}{1.2\,\text{V}}\right) = -\tau \ln 0 = \boxed{\infty}$$

6. **Task (c):** From 10 % to 90 % of $1.2\,\text{V}$:

$$\begin{aligned} t_{90\%} - t_{10\%} &= -\tau \ln(1 - 0.9) - (-\tau \ln(1 - 0.1)) \\ &= \tau \ln \frac{0.9}{0.1} = \tau \ln 9 \approx \boxed{2.75\,\text{ns}} \end{aligned}$$

> **Note:** These two results hold for **every** RC circuit and are used again later in the course: * 50 % delay: $t_{50\%} = \tau \ln 2 \approx 0.69\,\tau$ * 10 %–90 % rise time: $t_r = \tau \ln 9 \approx 2.2\,\tau$

Task 1.8

Idea: (a) is the same RC charging circuit as in Task 1.7. (b) is the discharging case: set up the differential equation with KCL and solve it the same way. Here $V_{DD} = 1.2\,\text{V}$, and V_{out} is the voltage across C_L.

1. **Task (a):** Charging via R_1 from $V_{out}(0) = 0$ to $V_{DD}/2$.
   - Time constant: $\tau_1 = R_1 C_L = 30\,\text{k}\Omega \cdot 1\,\mu\text{F} = 30\,\text{ms}$
   - Apply the formula from Task 1.7 with $V_{in} = V_{DD}$:

$$t_{PLH} = -\tau_1 \ln\left(1 - \frac{V_{DD}/2}{V_{DD}}\right) = \tau_1 \ln 2 = 30\,\text{ms} \cdot \ln 2 \approx \boxed{20.8\,\text{ms}}$$

2. **Task (b):** Discharging via R_2 from $V_{out}(0) = V_{DD}$ to $V_{DD}/2$.
   - The capacitor current flows out through R_2 (KCL):

$$C_L \frac{dV_{out}}{dt} = -\frac{V_{out}}{R_2} \quad \Rightarrow \quad dt = -R_2 C_L \frac{dV_{out}}{V_{out}}$$

   - Integrate both sides (integration variable v, from V_{DD} to V_{out}):

$$\int_0^t dt' = -R_2 C_L \int_{V_{DD}}^{V_{out}} \frac{dv}{v}$$

$$t = -\tau_2 \ln \frac{V_{out}}{V_{DD}}$$

   Solved for V_{out}: $V_{out}(t) = V_{DD}\, e^{-t/\tau_2}$.

- Time constant: $\tau_2 = R_2 C_L = 12.5\,\text{k}\Omega \cdot 1\,\mu\text{F} = 12.5\,\text{ms}$

$$t_{PHL} = -\tau_2 \ln \frac{V_{DD}/2}{V_{DD}} = \tau_2 \ln 2 = 12.5\,\text{ms} \cdot \ln 2 \approx \boxed{8.66\,\text{ms}}$$

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3. **Task (c):** Ratio of the two delays:

$$\frac{t_{PLH}}{t_{PHL}} = \frac{R_1 C_L \ln 2}{R_2 C_L \ln 2} = \frac{R_1}{R_2} = \frac{30\,\text{k}\Omega}{12.5\,\text{k}\Omega} = \boxed{2.4}$$

> **Note:** With the same load C_L, the ratio of the delays equals the ratio of the resistances. Later in the course, the pull-up network (PMOS) and pull-down network (NMOS) of a CMOS gate play the roles of R_1 and R_2. This is why the rise and fall delays differ, and why transistors are sized to make them equal.

Task 1.9

Idea: When the feature size scales by a factor S, both the length and the width of a transistor scale by S, so its **area scales by** S^2.

1. Scaling factor from 0.18 µm to 0.13 µm technology:

$$S = \frac{0.13\,\mu\text{m}}{0.18\,\mu\text{m}} \approx 0.7$$

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2. Area per transistor:

$$\frac{A_{0.13}}{A_{0.18}} = S^2 \approx 0.7^2 \approx 0.5$$

Each transistor needs only **half** the area, e.g. 50 million transistors need only about $0.7\,\text{cm} \times 0.7\,\text{cm} \approx 0.5\,\text{cm}^2$ instead of $1\,\text{cm}^2$.

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3. On the same chip area of $1\,\text{cm} \times 1\,\text{cm}$:

$$N_{0.13} = \frac{N_{0.18}}{S^2} \approx \frac{50 \cdot 10^6}{0.5} = \boxed{100\text{ million transistors}}$$

> **Note:** One technology generation scales by $S \approx 0.7$ and therefore **doubles** the number of transistors per area. This is Moore's law.

Task 1.10

Idea: With n technology generations, the clock frequency is multiplied by 2^n. Solve for n with the binary logarithm ld $= \log_2$.

1. Number of generations:

$$2\,\text{GHz} \cdot 2^n = 10\,\text{GHz}$$
$$2^n = 5$$
$$n = \text{ld}(5) = \frac{\ln 5}{\ln 2} \approx \boxed{2.3\ \text{generations}}$$

2. With 3 years per generation:

$$t = n \cdot 3\,\text{years} \approx 2.3 \cdot 3\,\text{years} \approx \boxed{6.9\ \text{years}}$$

Appendix: Boolean Algebra Laws

Law	OR form	AND form
Commutative	$A \vee B = B \vee A$	$A \wedge B = B \wedge A$
Associative	$A \vee (B \vee C) = (A \vee B) \vee C$	$A \wedge (B \wedge C) = (A \wedge B) \wedge C$
Distributive	$A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$	$A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$
Identity	$A \vee 0 = A$	$A \wedge 1 = A$
Null (Domination)	$A \vee 1 = 1$	$A \wedge 0 = 0$
Idempotent	$A \vee A = A$	$A \wedge A = A$
Complement	$A \vee \overline{A} = 1$	$A \wedge \overline{A} = 0$
Double negation	$\overline{\overline{A}} = A$	–
De Morgan	$\overline{A \vee B} = \overline{A} \wedge \overline{B}$	$\overline{A \wedge B} = \overline{A} \vee \overline{B}$
Absorption	$A \vee (A \wedge B) = A$	$A \wedge (A \vee B) = A$
Combining	$(A \wedge B) \vee (A \wedge \overline{B}) = A$	$(A \vee B) \wedge (A \vee \overline{B}) = A$